Cloud SDK for Scotty: Unified Resumable Upload Protocol

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Summary

Scotty (go/scotty, go/scotty-http-protocols) is a shared Google service for handling large HTTP/HTTPS uploads and downloads. Scotty acts as a frontend that offloads the complexity of file transfers from individual product backends. Google API teams using Scotty implement a "customer agent" backend that Scotty notifies about transfer progress and completion, allowing the team's application to manage the file and control access. Cloud SDK Libraries are adding support for Scotty functionality.

This document describes the Unified Resumable Upload Protocol from the Cloud SDK point of view. The Unified Resumable Upload protocol is designed and controlled by Google. The protocol description can be found here: Unified Upload Protocol / Resumable Upload Protocol. There is also a smaller overview: Scotty HTTP Protocols Overview / Unified Resumable Protocol. This document is not intended to repeat the protocol specification documents fully. Instead it presents the parts that are important for Client Libraries.

Note on terminology

Throughout this document I will refer to the Unified Resumable Upload protocol as simply "the upload protocol" or "the protocol".

I will talk about the protocol state machine, and generally use a state-machine related terminology. This does not mean that this document prescribes that the protocol should be implemented as a state machine. Rather, the protocol state machine is a concept used to simplify descriptions.

I will also use terms "phase" and "state" somewhat interchangeably. A protocol state machine being "in" a state "X" can be described as a protocol flow having the "X" phase: consisting of the state machine transitioning to state "X", performing some process inside that state (e.g. sending a request, retrying) and then transitioning out of the state "X".

The global deadline

The library invocation of the protocol session must have a global deadline. This deadline is separate from the:

- Local deadline, which is applied to any given HTTP request
- A deadline for Scotty server-side timeout for the upload session, which can be configured very large (up to 3 days) and allows the upload to be interrupted and continued later.

The deadline is configured as is usual for the Gapic-generated libraries:

- If the end-user is specified a deadline, that value takes precedence
- Then, if the API team has specified a deadline in grpc config, that value is taken
- Then, if neither the end-user nor the API team has specified a deadline, the Library configures a sensible default (e.g. 10 minutes).

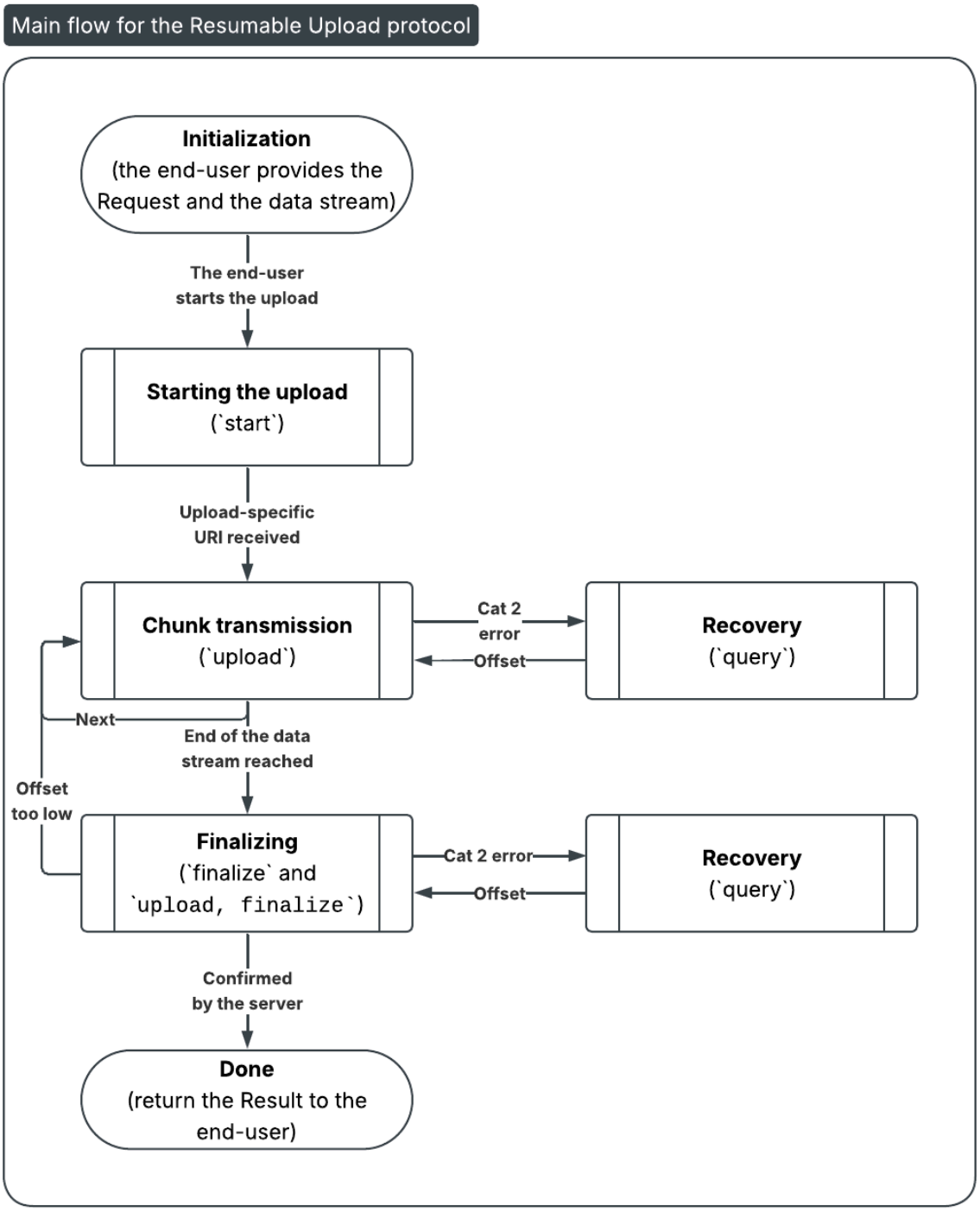
The sensible default might appear too large for the data transmission. Typical upload size is hundreds of megabytes, and those don't take 10 minutes to upload. However we configure this amount of time by default the libraries for even smaller calls because it has to account for server-side delays and retrying with backoff.

The only addition for Scotty is as following: the Client Libraries MAY revise the default deadline upward if:

(1) The end-user and the API team has not provided any values for the deadline (and so we have to use the default), and
(2) The end-user has provided the upload size (which is optional)

In that case the Client Libraries MAY use a sensible default for the upload speed (e.g. 100 Mb/s) and increase the deadline if the size divided by the upload speed default is greater than the default deadline. This is optional. We rely on the end-users to set the appropriate deadline for their uploads.

The main "Success" flow of the upload protocol



The typical successful protocol flow will consist of a "Starting" phase followed by one or more "Chunk transmission" phases, followed by a Finalizing phase. The finalizing phase will typically send a combined `upload, finalize` command so it will upload the last chunk and finalize the upload in the same request. In rare cases this phase will issue a separate `finalize` command.

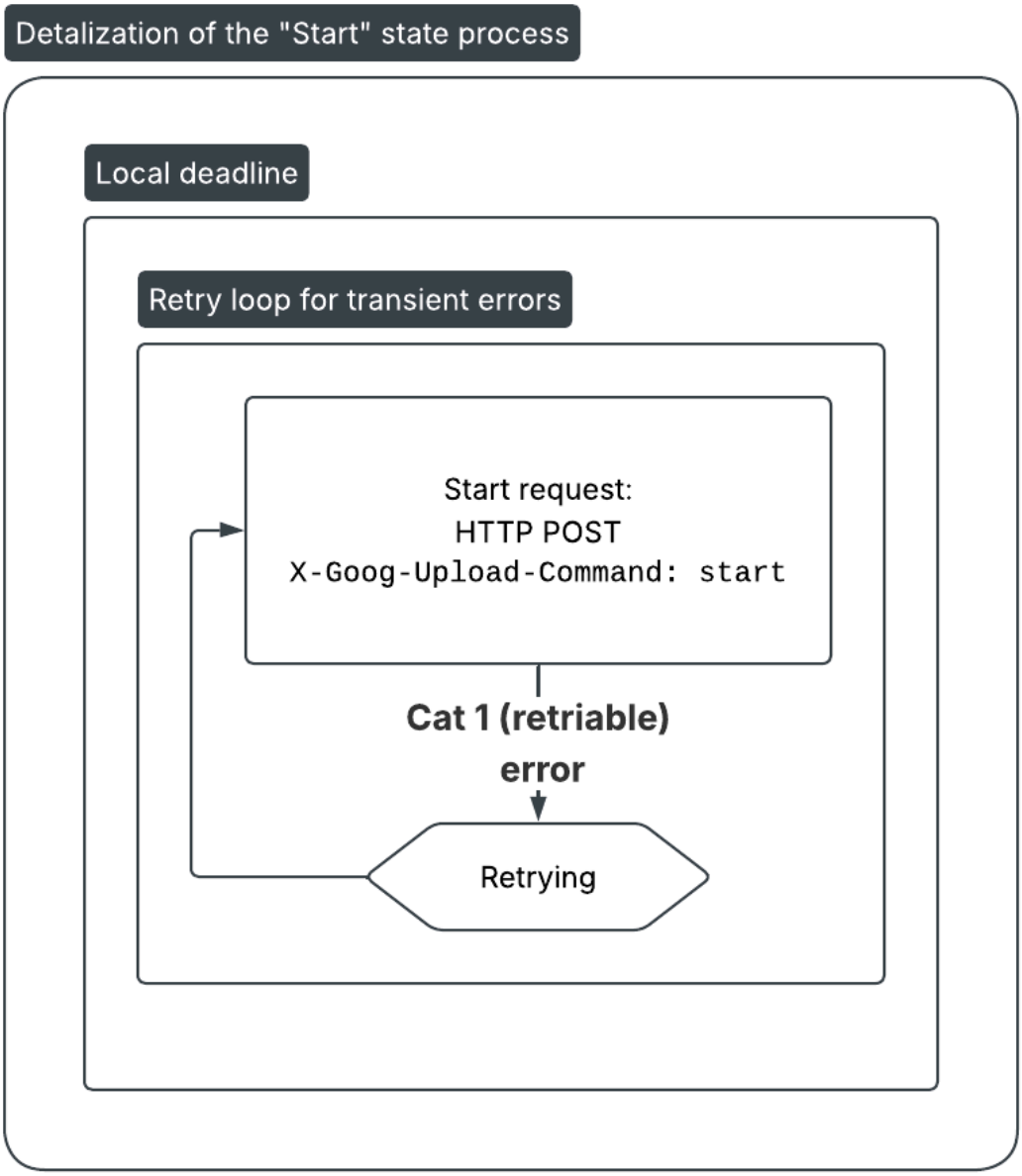
A note on the errors

This document refers to 3 categories of errors: Retriable (Cat 1), Recoverable (Cat 2), and Unrecoverable (Cat 3).

Category	**Description**	**Example Codes**	**Action**
Category 1	Retriable Without Modification (Transient)	`429`, `500`, `502`, `503`, `504`, TCP/Socket Timeout	Retry the request
Category 2	Retriable With Modification (State Mismatch)	`400`, `412`, `416`	Transition to the "Recovery" state
Category 3	Unrecoverable, Unretriable (Fatal)	`401`, `403`, `404`	Bubble up to the end-user

Phase structure

All the phases in the diagram above contain a retry with a deadline check. Here is a structure of a phase, using without loss of generality the "Start" phase as an example:



Every non-terminal state is associated with a protocol command that is sent as a HTTP request to the server-side endpoint. For the "Chunk transmission" state the command is `upload`. Every HTTP request must be wrapped in a retry loop with a local deadline.

The retry loop must retry only retriable errors. The retriable error that has exhausted either the retry timeout or attempt count becomes a terminal error.

The local deadline

The local deadline is distinct from the global deadline specified for the upload attempt as a whole, although they can be equal, especially towards the end of the long upload session. The recommendation from the BQ team was to set the local deadline to be at most the length of the half of the initial timeout. E.g. if the end-user has specified the initial deadline to be 10 minutes from the start of the call, then the initial timeout is 10 minutes, and the local deadlines should have a length of at most 5 minutes from the beginning of the call. This, however, is left to the language implementations.

"Starting" state

The command for the "Starting" phase is `start`. It is very distinct from other Scotty commands.

Different "Starting" endpoint and the upload prefix

It will be issued to a different endpoint from the other commands. The "Starting" endpoint is constructed from the endpoint specified in `google.api.http` annotation with the addition of the scotty upload prefix, which is typically `resumable/upload`. The API team members will provide the upload prefix when opting-in to the Scotty generation, as part of the configuration in the libraries section of the services config.

Example:

```
None
# In the API protos
rpc CreateYouTubeVideoUpload(CreateYouTubeVideoUploadRequest)
      returns (CreateYouTubeVideoUploadResponse) {
    option (google.api.http) = {
      post: "/v24/customers/{customer_id=*}/youTubeVideoUploads:create"
  }
}

# In the libraries generation configuration and in the library code
Configured upload prefix: `/resumable/upload`.

# At runtime:
"Starting" url is calculated to be
"https://{host}/resumable/upload/v24/customers/foo/youTubeVideoUploads:create"
instead of
"https://{host}/v24/customers/foo/youTubeVideoUploads:create"
```

The end-user provided headers

The end-user must be able to specify headers that will be sent with the `start` command only. An example of such a header is the Google Ads developer key header `developer-token`.

The RequestMessage as the body

The request message of the Scotty RPC will be sent as a body of the `start` command (serialized in JSON). It will typically contain the upload data's semantic metadata such as the video title.

The upload URL

The response to the `start` command will contain the Scotty upload URL which contains the Scotty upload ID. The upload ID must be surfaced to the end-user. The upload URL can be configured (in the Scotty server configuration) to be completely arbitrary, so we cannot guarantee that we can parse the upload ID. The URL will have to be surfaced in its entirety.

The server-specified chunk size

In response to the `start` command, the server might specify a required chunk size with `X-Goog-Upload-Chunk-Granularity` header, e.g. `X-Goog-Upload-Chunk-Granularity: 50`. The client must always upload in the multiples of specified granularity, typically by changing its chunk size to be a closest smaller multiple of the server-specified granularity to the chunk size specified by the end-user (or the default).

Successful start of the upload (`X-Goog-Upload-Status: active`)

In response to the successful `start` command the server sends `X-Goog-Upload-Status: active` in the headers. This header should always be present in server responses during the protocol exchange, until it changes to `X-Goog-Upload-Status: final` after the successful "Finalizing" phase.

The absence of this header is an error: recoverable for "Uploading" and "Finalizing" state, and retriable for all other states.

Note, that there are two other possible states for this header:

- Receiving a response with `X-Goog-Upload-Status: final` in a response with a non-200 HTTP code is a rejection.
- It is expected that the value of the header will be `X-Goog-Upload-Status: cancelled` after a successful cancellation.

"Start" state transitions

After the successful `start` command, the protocol transitions to the `Chunk transmission` state.

In response to errors, the `start` command can only be retried. Since the upload session has not been created, there is no upload URL and the recovery is not possible. Similarly, the end-user cannot `cancel` when the `start` command is ongoing, so the only other transitions possible are to the terminal "Rejected" or "Error" states.

"Transmission" state

The command for the "Transmission" phase is `upload`. During a successful upload session this state repeats until all the chunks are successfully uploaded (the end-user provided data stream reaches EOF).

Transition to the Recovery state

In this phase the protocol implementation can encounter recoverable (aka "Category 2", aka "retriable with modification") errors. The main example of these is `HTTP 416 Range not satisfiable` . These errors indicate a disagreement between the server and the client on the upload offset. When such an error occurs the protocol implementation should transition to the "Recovery" phase.

Transition to the Finalizing state

This transition does not occur after analyzing the server response to the HTTP request. Instead it occurs when reading from the end-user provided data stream and reaching the end of it. When that happens typically there will be some amount of data just read, but it might happen that there is no data. It makes no difference to this state transition, the data or lack thereof is passed along.

"Finalizing" state

The command for the "Finalizing" phase is `finalize`.

The combined `upload, finalize` command.

If there is no data left over from the "Transmission", the body is sent empty with this command. In most cases though, there will be some data, typically not a multiple of a chunk size, that needs to be sent along. That data must be sent with `upload, finalize` combined command and not with the `upload` command. There are two concerns here:

First, if a server has chosen a granularity the data of the wrong size will be rejected.

Second, sending a combined command saves a roundtrip compared to sending two commands.

For these reasons we transition to the "Finalizing" state on end-of-stream instead of first sending the data in the "Transmitting" state.

Successful completion of the upload (`X-Goog-Upload-Status: final`)

After receiving the `finalize` command or the combined `upload, finalize` command, the server acknowledges the successful completion of the upload session by sending a response with the HTTP 200 code and the `X-Goog-Upload-Status: final` header. Only at that point the upload is considered successful.

NB: if the `X-Goog-Upload-Status: final` header is sent with any other status code it's a rejection instead.

Transition to the Recovery state

In this phase, just like in the "Transmission" phase the protocol implementation can still encounter recoverable (aka "Category 2", aka "retriable with modification") errors. Similarly to the "Transmission" phase" the protocol implementation should transition to the "Recovery" phase. These errors should not occur in response to the `finalize` command since all the data should be committed by that point. However if they occur, there is no issue with making a state transition to "Recovery".

The `finalize` command

The above means that there are two sets of circumstances where a `finalize` command is issued:

(1) The end-user-provided data object was an exact multiple of the selected chunk size. After the final chunk of the data was transmitted, the next stream read returned 0 bytes of data and an end-of-stream indicator.
(2) There was a recoverable error during the "Finalizing" phase. After the recovery, the server-specified offset was at the exact end of the end-user-provided data object (meaning all the data was successfully committed by the server before the recoverable error was returned).

Transition back to the "Transmission" state

It is possible that, during the recovery, the server replies with an offset that is lower than the offset of the data currently in the buffer of the "Finalizing" state. Theoretically if the server has discovered that the data is lost, it might return 0 requiring the data to be sent anew. In this case the "Finalizing" state should perform a transition back to the "Transmission" state. There are two operating concerns here:

First, if the end-user has specified the data granularity we should not send the amount of data greater than that.

Second, reading from and rewinding the end-user-provided data stream has some special logic (not covered in this document) and it's easier if that logic is kept contained in one state.

"Recovery" state

The command for the "Recovery" state is `query`. This command can technically be issued at any time, and in parallel with any other command. However we should not expose this functionality to the end-user; instead the command should only be issued in the "Recovery" state.

Reaching the "Recovery" state

This state should be transitioned to when a recoverable error is encountered during the "Transmission" or "Finalizing" state. The recoverable errors are also known as the Cat 2 errors, or errors retriable with modification. The purpose of the "Recovery" state is to obtain the information for this modification – the offset of the data that the server has committed. After this information is transmitted back to the "Transmission" or "Finalizing" state, those states can prepare a new chunk of data.

Why have the "Recovery" state

Recovery state is a part of the protocol specification. If a server error indicates chunk misalignment a blind retry is only going to be met with the same error. And since the server does not indicate in an error what the correct offset is we should issue the `query` command – which is the "Recovery" state.

Why have retries when there is the "Recovery" state

For the `upload` command in the "Transmission" state and the `upload, finalize` command in the "Finalizing" state it is possible to treat every retriable error as recoverable instead and issue a `query` command before continuing. I think this is excessive. Certain errors like `503 Unavailable` and `504 Gateway timeout` typically come from GFE and do not require any request modification (and Scotty documentation in the table here adds `408 Request Timeout` and `429 Too many requests` to that list). I don't think we will achieve some large simplification benefit by avoiding retries either since the retries are handled by the existing retry primitives in the same Gax library.

If a server committed a chunk of data before the error happened it will reply to a retry request with a recoverable error and then we can enter recovery so the retry is not going to put the protocol implementation in a terminal state either.

Counting Recovery attempts

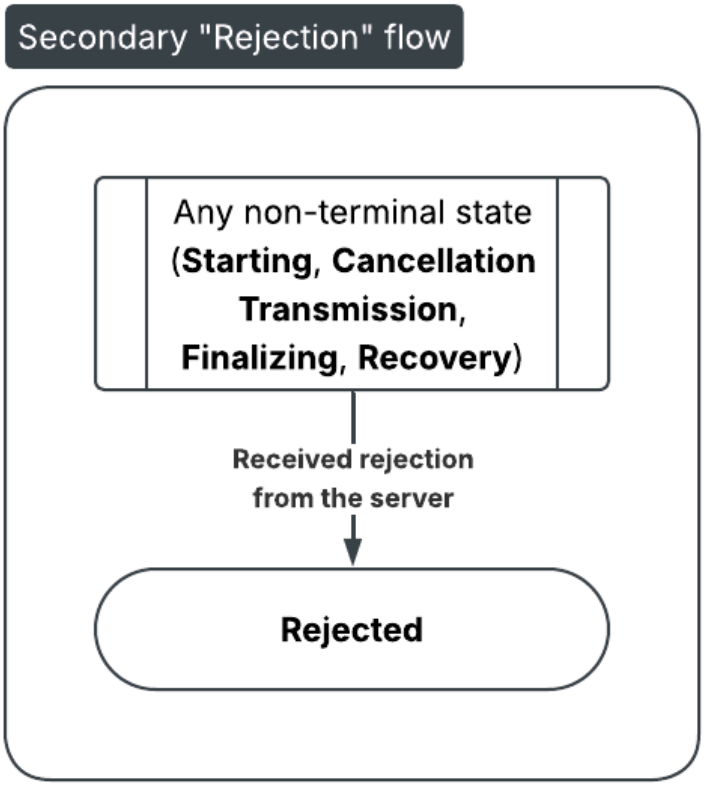
To avoid bouncing repeatedly between `upload` and `query` in case of an issue where an error is unrecoverable but the server returns a recoverable error code, the libraries must not attempt to retransmit after recovery more than 3 times **with the same offset**. In other words, if the chunk commands return a recoverable error, and the query commands return the same offset the upload session should be terminated with an unrecoverable error.

Secondary ("Unsuccessful") flows

The above documentation covered the states, commands, and transitions during the successful upload session, which might include a recovery. There are, however, a number of secondary flows that lead to different terminal outcomes and will not occur during the successful upload.

Rejection flow

For any command, the server may return the header `X-Goog-Upload-Status: final` in a response with a non-200 HTTP code. This means that the server has rejected the upload. The reason for this will be given in the body of the response and should be raised like a typical server error to the end-user.

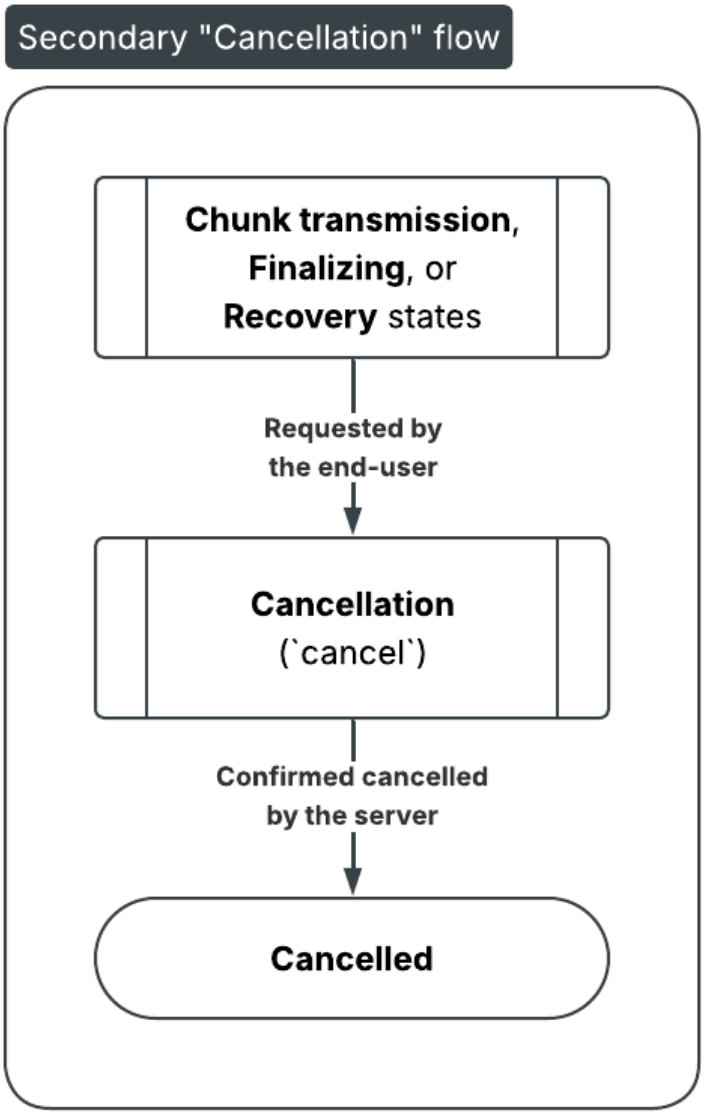


Note that this flow does not have any new non-terminal states.

Cancellation flow

At any point after the start of the upload and before the upload is finalized, the end-user should be able to request a cancellation. The mechanism of that is language-specific; one proposed option is to utilize the callback function that would notify the end-user of the progress.

This flow has a special State: Cancellation.



"Cancellation" state

The command associated with the "Cancellation" state is `cancel`. This command must be retried if it fails with a Cat 1 error. The successful cancellation of the upload will result in the server sending the response with the `X-Goog-Upload-Status: final`, after which the end-user must be notified. The mechanism is left for the language-specific implementation.

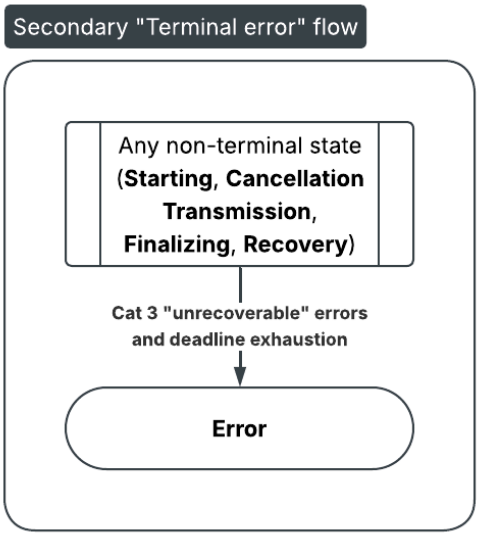
Not cancelling during cancellation

The protocol implementation should not give the end-user an opportunity to request cancellation during the "Cancellation" phase and must not attempt to make a cancellation request while another cancellation request is not done.

Terminal error flow

Certain errors are non-retriable and non-recoverable. This includes:

- Being in a recoverable state (e.g. "Transmission") and receiving an error that is not retriable or recoverable
- Being in a non-recoverable state (e.g. "Starting") and receiving an error that is not retriable
- Retrying a retriable error, and exhausting either the local deadline or attempt count
- Exhausting a global deadline



Note that this flow does not have any new non-terminal states.

The protocol state transition tables

This table includes states that are important **for the implementation**. Specifically the "Transmission" state and the "Finalizing" state have been split into sub-states for reading data and sending data. This document is describing a protocol and as such does not put reading data from the stream into focus. I did not however want to have two mostly similar state transition tables, so the implementation-scoped table had to appear in the protocol-scoped document.

The protocol state transition table: Success flow

From state		Event	To state	
Terminal	*Non-terminal*		*Non-terminal*	*Terminal*
Initializing		`start_upload` called	Starting	

	Starting	Response: 200, X-Goog-Upload-Status: active	Transmission | Reading from stream	
	Transmission | Reading from stream	Chunk-size data read, no EOF	Transmission | Sending	
	Transmission | Reading from stream	Any size data read, EOF	Finalizing | Reading from buffer	
	Transmission | Sending	Recoverable error	Recovery | From Transmission	
	Transmission | Sending	Response: 200, X-Goog-Upload-Status: active	Transmission | Reading from stream	
	Finalizing | Reading from buffer	Offset within buffer, Buffer contain data	Finalizing | Sending with upload	
	Finalizing | Reading from buffer	Offset within buffer, Buffer contain no data	Finalizing | Sending finalize	
	Finalizing | Sending with upload	Recoverable error	Recovery | From Finalizing	
	Finalizing | Sending with upload	Response: 200, X-Goog-Upload-Status: final		**Success**
	Finalizing | Sending finalize	Recoverable error	Recovery | From Finalizing	
	Finalizing | Sending finalize	Response: 200, X-Goog-Upload-Status: final		**Success**
	Recovery | From Transmission	Response: 200, X-Goog-Upload-Status: active, Offset	Transmission | Reading from stream	
	Recovery | From Finalizing	Response: 200, X-Goog-Upload-Status: active, Offset	Finalizing | Reading from buffer	

The protocol state transition table: Unsuccessful flows

From state		**Event**	**To state**	
Terminal	*Non-terminal*		*Non-terminal*	*Terminal*
	Transmission | Sending	End-user event: `cancel`	Cancelling	
	Finalizing | Sending with upload	End-user event: `cancel`	Cancelling	
	Finalizing | Sending finalize	End-user event: `cancel`	Cancelling	
	Recovery | From Transmission	End-user event: `cancel`	Cancelling	
	Recovery | From Finalizing	End-user event: `cancel`	Cancelling	
	Cancelling	Response: 200, X-Goog-Upload-Status: cancelled		**Cancelled**
	Starting	Category 3 "Unrecoverable" errors		**Error**
	Transmission | Reading from stream	Category 3 "Unrecoverable" errors		**Error**
	Transmission | Sending	Category 3 "Unrecoverable" errors		**Error**

	Finalizing | Reading from buffer	Category 3 "Unrecoverable" errors		**Error**
	Finalizing | Sending with upload	Category 3 "Unrecoverable" errors		**Error**
	Finalizing | Sending finalize	Category 3 "Unrecoverable" errors		**Error**
	Recovery | From Transmission	Category 3 "Unrecoverable" errors		**Error**
	Recovery | From Finalizing	Category 3 "Unrecoverable" errors		**Error**
	Cancelling	Category 3 "Unrecoverable" errors		**Error**
	Starting	Rejection (Response: non-200, X-Goog-Upload-Status: final)		**Rejected**
	Transmission | Sending	Rejection (Response: non-200, X-Goog-Upload-Status: final)		**Rejected**
	Finalizing | Sending with upload	Rejection (Response: non-200, X-Goog-Upload-Status: final)		**Rejected**
	Finalizing | Sending finalize	Rejection (Response: non-200, X-Goog-Upload-Status: final)		**Rejected**
	Recovery | From Transmission	Rejection (Response: non-200, X-Goog-Upload-Status: final)		**Rejected**
	Recovery | From Finalizing	Rejection (Response: non-200, X-Goog-Upload-Status: final)		**Rejected**
	Cancelling	Rejection (Response: non-200, X-Goog-Upload-Status: final)		**Rejected**